

# Perceptual Visibility Prediction of Overlaid Content in Optical See-Through Augmented Reality

光学シースルー拡張現実における  
重畳コンテンツの知覚的視認性予測

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# Outline

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1. Introduction
2. Related Work
3. Method
4. Pilot Experiment
5. Conclusion & Future Work

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- 1. Introduction** — background, scope, objective
2. Related Work
3. Method
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# Introduction: Optical See-Through AR

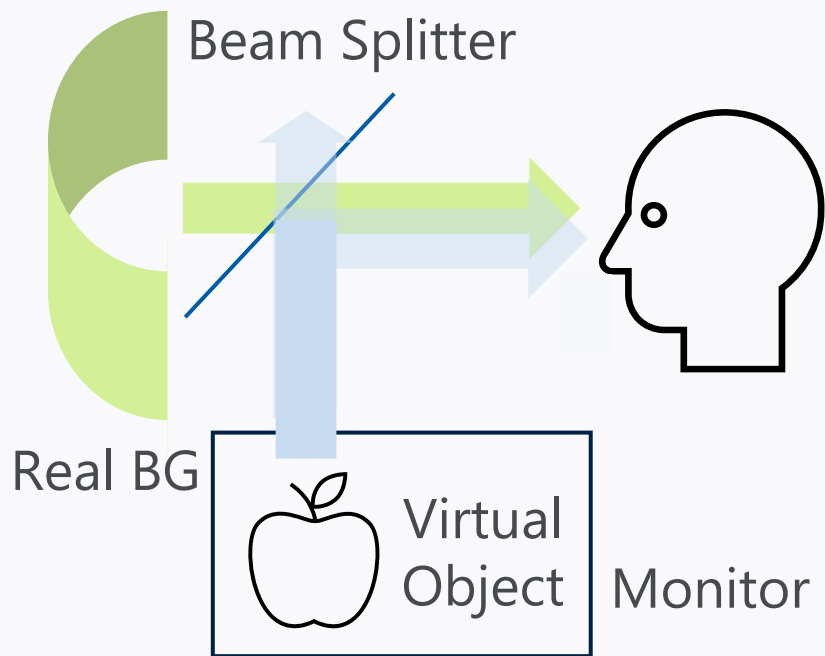
A virtual image is **optically overlaid** on the real-world background

## Pros:

Direct perception of real world

## Cons:

**Additive mixture** of FG & BG



# Introduction: Visibility Depends on the Background

**Visibility of overlayed content varies** depending on backgrounds



**High visibility**



**Low visibility**  
→ **Bad UX, Safety**

→ Need to **predict visibility** from the FG/BG image

# Scope & Objective

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**Goal:** Build an **image-based visibility predictor** for OST-AR

## Three steps

1. **Psychophysical Experiment :** Reveal HVS mechanism
2. **Model Extension:** From 2D to OST-AR
3. **Model Validation:** model's predictions vs human ratings

# Outline

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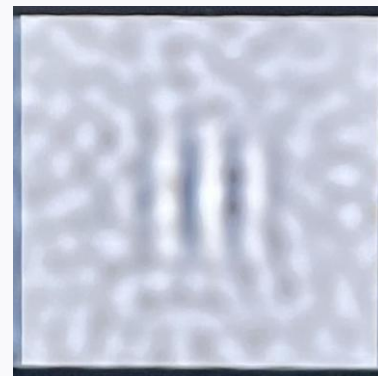
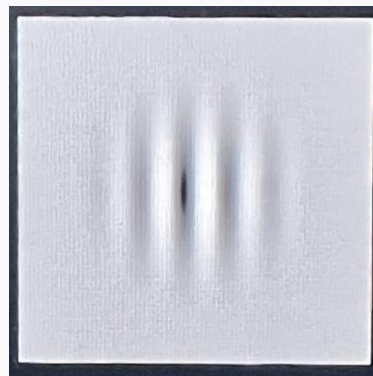
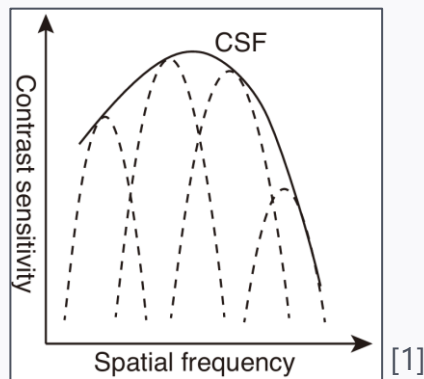
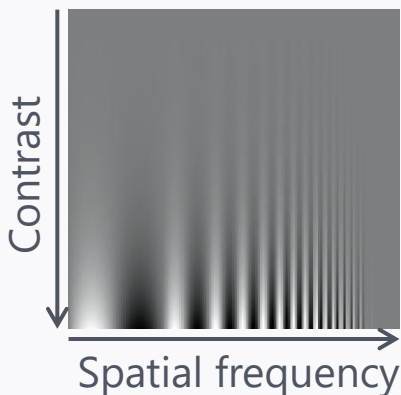
1. Introduction
2. **Related Work** — four topics
3. Method
4. Pilot Experiment
5. Conclusion & Future Work

# Related Work (1): Contrast Perception

Visibility ← **Contrast Perception** | Two phenomena addressed

① Frequency dependence (CSF)

② Contrast masking



→ This work mainly focus on ② **contrast masking**



## Related Work (2): Single-Factor Models for OST-AR

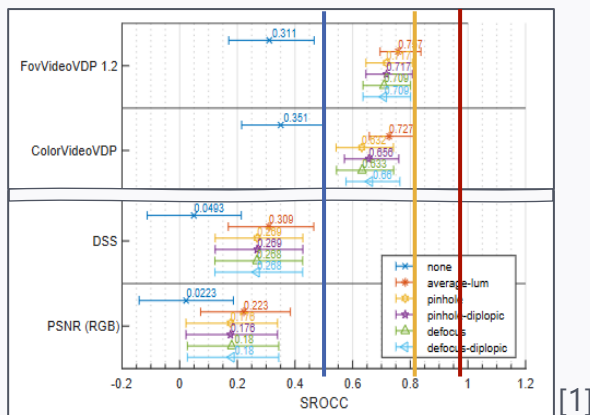
Existing models each consider **one perceptual factor**

| Perceptual factor            | Reference                                   |
|------------------------------|---------------------------------------------|
| <b>Color</b>                 | Yoon et al., 2017 · Zhang et al., TVCG 2022 |
| <b>Luminance/Illuminance</b> | Liu et al., ISMAR 2022                      |
| <b>Transparency</b>          | Zhang & Murdoch, 2021                       |

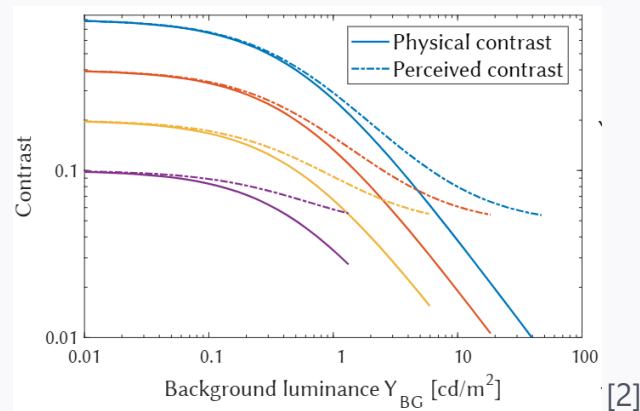
→ **No unified, multi-factor model exist**

# Related Work (3): IQA for OST-AR Visibility

IQA closely relates to visibility — also in **existing-model limits**



Limitation of existing  
IQA models



OST-AR specific  
contrast perception

→ **OST-AR-specific perception must be modeled.**

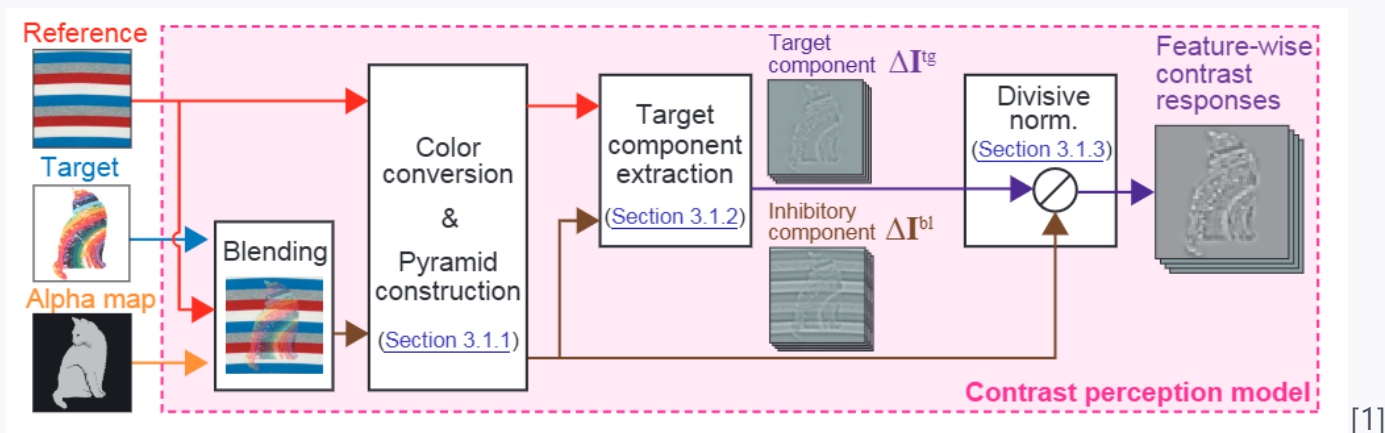
[1] Chapiro, et al. "AR-DAVID: Augmented Reality Display Artifact Video Dataset." 2024

[2] Kim et al. "Supra-threshold Contrast Perception in Augmented Reality." 2025

## Related Work (3): Fukiage's Visibility Predictor

Visibility predictor for  **$\alpha$ -blended content** — promising for OST-AR

! Considers **multiple perceptual factors** jointly



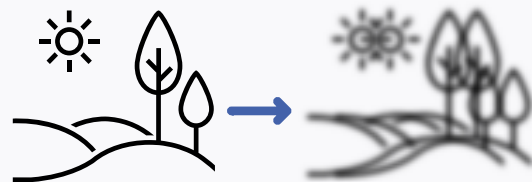
→ But **some gaps** remain when applied to OST-AR

## Related Work (4): Fukiage's Visibility Predictor

The model assumes  **$\alpha$ -blending** — single 2D display

Four OST-AR specific gaps:

1. **Accommodation:** induces defocus blur
2. **Binocular disparity:** arises with depth difference
3. **BG discounting:** observers partly ignore the real BG
4. **Optical addition:** HDR-range output



→ This work **bridges these gaps**, extending the model to OST-AR

# Outline

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1. Introduction
2. Related Work
3. **Method** — overview & three steps
4. Pilot Experiment
5. Conclusion & Future Work

# Method: Overview

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**Three steps** to extend Fukiage's model for OST-AR

## **STEP 1** Visibility Evaluation Experiment

— Quantify perceptual effects on visibility

## **STEP 2** Model Extension

— Extend input processing from results

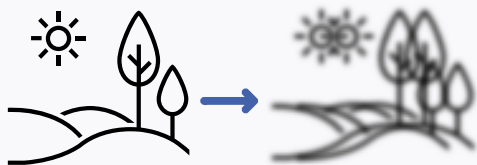
## **STEP 3** Model Validation

— Compare with human rating

# Method (1): Visibility Evaluation Experiment

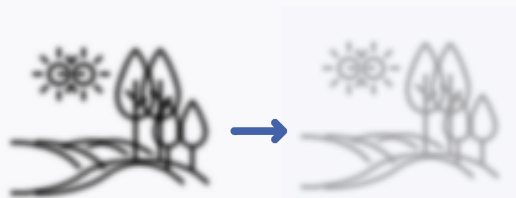
Quantify **how each OST-AR specific factor affect visibility**

## BG perception change



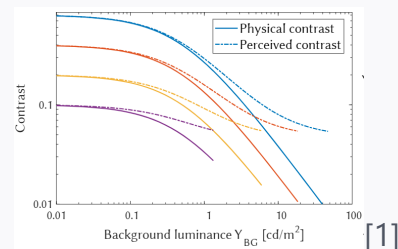
**Defocus  
+ Binocular disparity**

## BG discounting



**Non-optical  
perception change**

## Linear summation



**Contrast perception  
change**

→ **Decide extensions & fit parameters**

# Method (2): Model Extension

Extend model pipeline with **four OST-AR-specific mechanisms**

## ① **Defocus blur** (Calculate PSF)

$$h(\vec{u}; \Delta D, p) = \begin{cases} 1, & \|\vec{u}\| < p\Delta D/2 \\ 0, & \text{otherwise.} \end{cases}$$

## ② **Disparity & combination**

Geometric computation  
& existing-model selection

## ③ **HDR → SDR** (Tone mapping)

$$L_d(x) = \frac{L(x)}{1 + L(x)} \quad [1]$$

## ④ **Background discounting**

Fit the attenuation process



## Method (3): Model Validation

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Evaluate the model by **comparing with human evaluation**

### Psychophysical experiment

Visibility evaluation in OST-AR, using **general texture images**.



### Model Evaluation

Quantitatively compare **human scores** with **model predictions**  
(e.g.,  $R^2$ , SROCC)

# Outline

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1. Introduction
2. Related Work
3. Method
4. **Pilot Experiment** — design, results, discussion
5. Conclusion & Future Work

# Pilot Experiment: Objective

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Identify the **parameters & conditions** for the main experiment

Quantify how **defocus blur** & **uncrossed disparity** affect visibility, across:

## Stimulus parameters

- Spatial frequency
- FG / BG luminance
- BG contrast

## Viewing conditions

- Diopter difference
- Ocularity

# Pilot Experiment: Apparatus

## Optical components

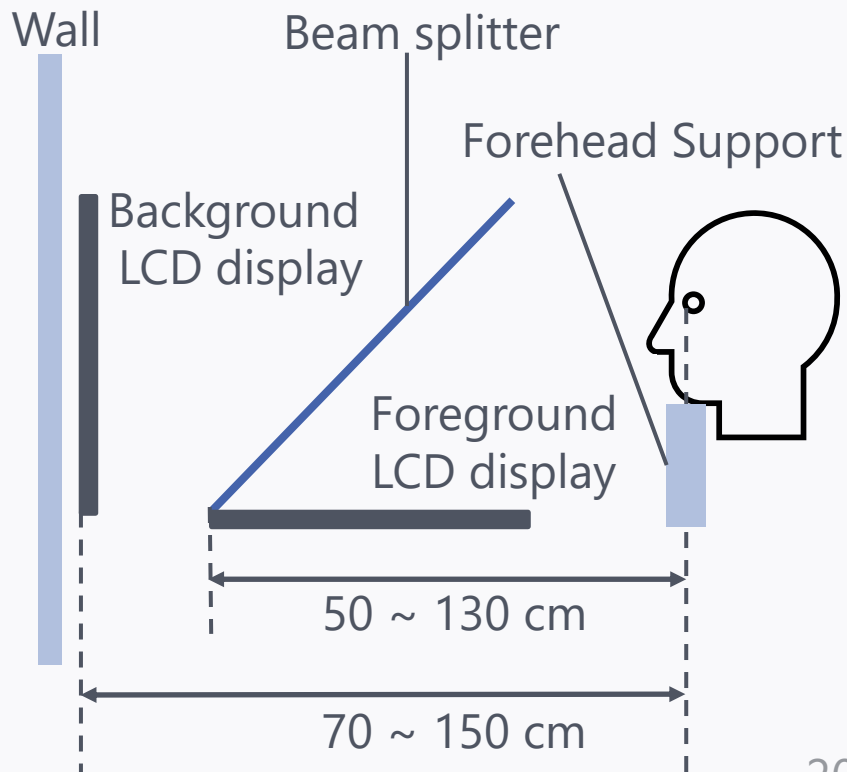
- Beam splitter:  
T 20–30%, R 50–60%
- $2 \times 27''$  LCD,  $2560 \times 1440$

## Luminance

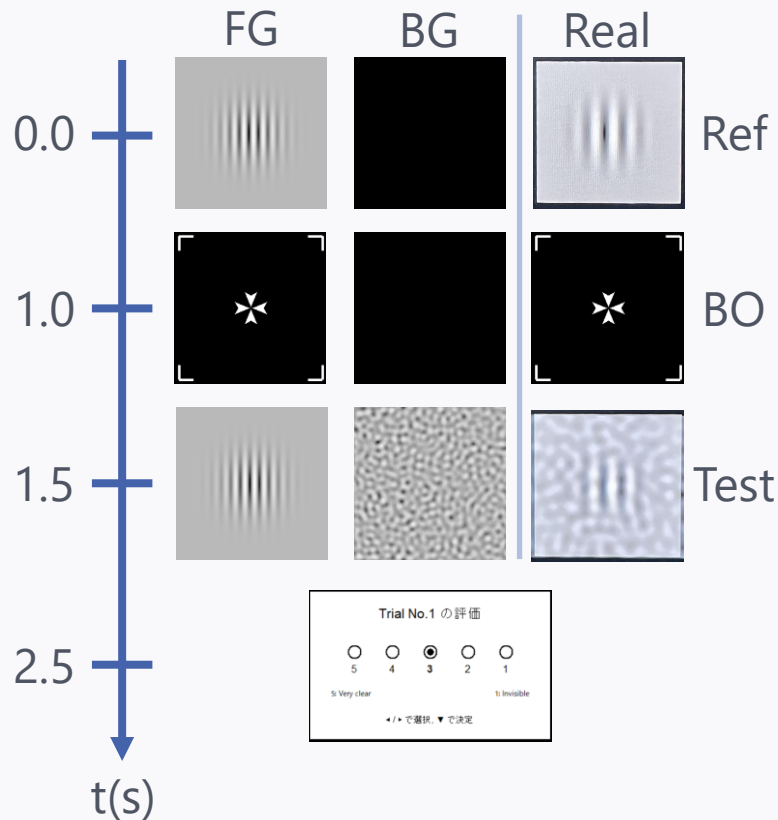
- FG max:  $100 \text{ cd/m}^2$
- BG max:  $30 \text{ cd/m}^2$

## Viewing geometry

- Chin rest, dark room



# Pilot Experiment: Procedure



## Rating scale

- **1** Invisible — **5** Very clear

## Stimulus pairing

- **Reference:** Gabor patch
- **Test:** Gabor + BG noise

## Trials

- All combinations  $\times$  2 repeats
- Randomized
- $N = 1$  observer (author)

# Pilot Experiment: Stimuli

| Specification                              | Value                                             |
|--------------------------------------------|---------------------------------------------------|
| Stimuli size                               | $7.9^\circ \times 7.9^\circ$                      |
| Envelope of Gabor patch ( $\sigma$ )       | $1^\circ \times 1^\circ$                          |
| Spatial frequency (SF, cpd in Gabor patch) | 2, 4, 6, 8 cpd                                    |
| Foreground luminance( $Y_{FG}$ )           | <u>50 cd/m<sup>2</sup></u> or 5 cd/m <sup>2</sup> |
| Background luminance( $Y_{BG}$ )           | 15 cd/m <sup>2</sup> or 5 cd/m <sup>2</sup>       |
| Foreground contrast( $C_{FG}$ )            | 1.0                                               |
| Background contrast( $C_{BG}$ )            | 1.0 or 0.0                                        |

# Pilot Experiment: Conditions

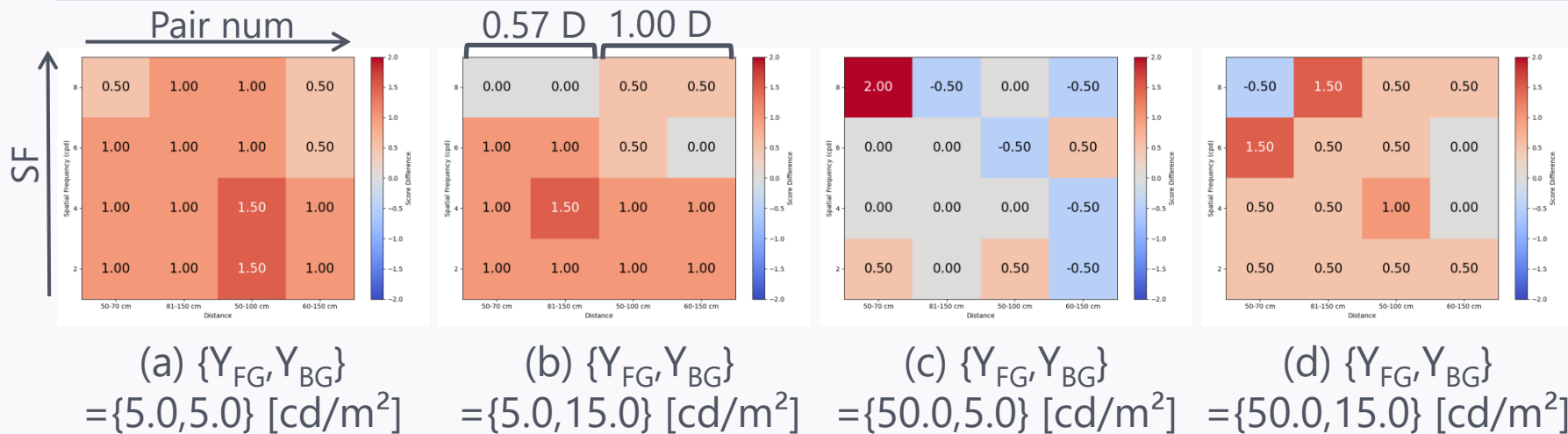
| Pair num | $d_{FG}$ [m] | $d_{BG}$ [m] | $D_{FG}$ [D] | $D_{BG}$ [D] | $\Delta D$ [D] |
|----------|--------------|--------------|--------------|--------------|----------------|
| 1        | 0.50         | 0.70         | 2.00         | 1.43         | 0.57           |
| 2        | 0.81         | 1.50         | 1.23         | 0.67         | 0.57           |
| 3        | 0.50         | 1.00         | 2.00         | 1.00         | 1.00           |
| 4        | 0.60         | 1.50         | 1.67         | 0.67         | 1.00           |

✂ Defocus blur arise **over 0.3 D**

## Other conditions

- Monocular (no disparity) vs Binocular (disparity & combination)

# Pilot Experiment: Results — Monocular

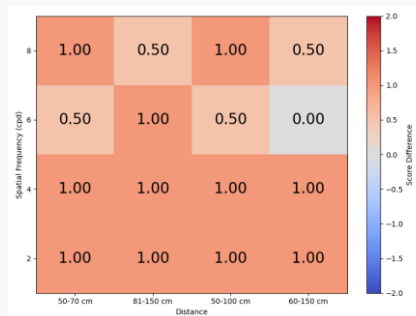


## Key observations

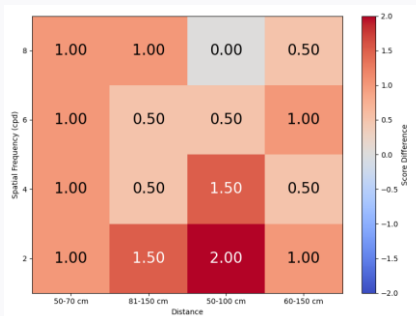
- Strongest degradation at (a)(b)
- No consistent  $\Delta D$  trend
- Low SF (2–4 cpd) more affected
- Some negative scores



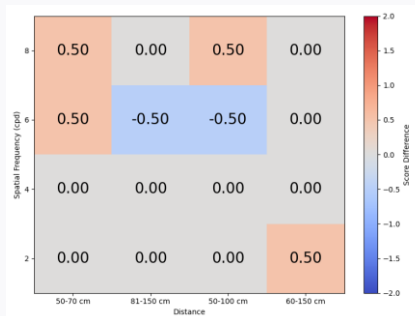
# Pilot Experiment: Results — Binocular



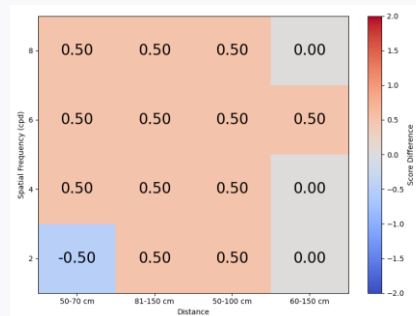
(a)  $\{Y_{FG}, Y_{BG}\}$   
 $=\{5.0, 5.0\}$  [cd/m<sup>2</sup>]



(b)  $\{Y_{FG}, Y_{BG}\}$   
 $=\{5.0, 15.0\}$  [cd/m<sup>2</sup>]



(c)  $\{Y_{FG}, Y_{BG}\}$   
 $=\{50.0, 5.0\}$  [cd/m<sup>2</sup>]



(d)  $\{Y_{FG}, Y_{BG}\}$   
 $=\{50.0, 15.0\}$  [cd/m<sup>2</sup>]

## Key observations

- Luminance, SF, negative scores patterns same as monocular
- No consistent  $\Delta D$  trend
- No mono-bino difference

# Pilot Experiment: Discussion

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## Four redesign directions for the main experiment

### ① Drop weak masking conditions

**Discrete rating lacks resolution**

where the true effect is small

### ② Fix luminance & spatial freq

**High  $Y_{BG}$ , mid  $Y_{FG}$ , and low SF**

→reduce other axes

### ③ Switch to matching method

Use **slower, but more accurate**

method than discrete rating

### ④ Add single-plane condition

Simulate

**linear addition & defocus blur**

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# Conclusion & Future Work

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## Conclusion

- **Goal:** extend Fukiage's visibility predictor to OST-AR through an experiment
- **Pilot experiment:** design study for that experiment
- **Pilot result:** identified the key factors and parameter ranges to target

## Future Work

**Add pre-experiment**  
to check redesign



**Main experiment**  
after ethical review



**Model Extension  
& Validation**